

MS Data Engineering

Module 1 — Part I

Time-Variant Data and Classical Time Series Analysis

Topics covered:

- Nature and challenges of temporal data
- Stationarity, autocovariance, autocorrelation
- Trend, seasonality, and classical decomposition
- ARIMA and Seasonal ARIMA models
- Vector Autoregression (VAR)
- State-space models and the Kalman Filter

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1 Nature and Challenges of Temporal Data

1.1 What Distinguishes Time Series Data

A **time series** is a sequence of observations indexed by time:

$$\{y_t\}_{t=1}^T \subset \mathbb{R}^d,$$

where $d = 1$ is the univariate case and $d > 1$ the multivariate case. Unlike a typical machine-learning dataset of i.i.d. samples, temporal data violates the independence assumption in a fundamental way: the value y_t carries information about future values y_{t+h} for $h \geq 1$. This temporal dependence is not a nuisance to be corrected—it is the primary source of information for forecasting.

Central challenge of time series

Standard statistical inference assumes $y_1, y_2, \dots, y_T \stackrel{\text{iid}}{\sim} F$. Time series analysis drops the independence assumption and replaces it with a *model of dependence across time*. Almost every classical and modern technique in this module is, at its core, a way of specifying and exploiting that dependence structure.

1.2 Taxonomy of Temporal Patterns

Definition 1.1 (Trend)

A long-run systematic movement in the mean level of the series. Trends may be *deterministic* (e.g. $\mu_t = \alpha + \beta t$) or *stochastic* (a random walk with drift).

Definition 1.2 (Seasonality)

A pattern that repeats with a *known, fixed period* s : $\mathbb{E}[y_t] = \mathbb{E}[y_{t+s}]$ for all t . Examples: daily temperature ($s = 24$ hours), monthly retail sales ($s = 12$ months), weekly web traffic ($s = 7$ days).

Definition 1.3 (Cycles)

Quasi-periodic fluctuations with a *variable or unknown* period, often driven by economic or ecological mechanisms. Unlike seasonality, cycles are not strictly periodic and cannot be removed by a simple seasonal adjustment.

Definition 1.4 (Irregular component)

The residual after trend, seasonality, and cycle have been accounted for. Ideally this is white noise; if not, the model has missed structure.

1.3 Challenges Specific to Temporal Data

1. **Non-stationarity.** Most raw series exhibit evolving mean, variance, or autocorrelation. Classical inference assumes stationarity; failure to achieve it leads to spurious results.
2. **Multiple seasonalities.** High-frequency data (hourly electricity demand, call-centre traffic) exhibit daily *and* weekly seasonalities simultaneously, which ARIMA cannot handle natively.

3. **Irregular sampling and missing values.** Sensor outages, trading halts, and survey gaps create irregular grids that invalidate standard autocorrelation estimators.
4. **Temporal leakage in model evaluation.** Randomly splitting a time series into train/test sets violates temporal order and inflates performance estimates. Proper evaluation always uses *forward-chaining* (walk-forward) cross-validation.
5. **Non-linearity and regime changes.** Financial crises, pandemics, and policy shifts create structural breaks that violate stationarity and make long-horizon forecasting unreliable.
6. **High dimensionality.** Modern data engineering produces thousands of correlated time series (e.g. server metrics, sensor arrays). Joint modelling is essential but computationally demanding.

Example 1.1 (Temporal leakage)

You have 5 years of daily stock returns. A naive 80/20 train/test split assigns random days to the test set. A model trained on data from 2024 and tested on a random draw including 2022 data has *seen the future* during training. The correct approach: train on years 1–4, test on year 5.

2 Stationarity and the Autocorrelation Function

2.1 Stochastic Processes

Formally, a time series is a realisation of a *stochastic process*: a family $\{Y_t, t \in \mathbb{Z}\}$ of random variables defined on a common probability space $(\Omega, \mathcal{F}, P)$. A single observed series $\{y_t\}_{t=1}^T$ is one realisation (path) of this process.

Definition 2.1 (Strictly stationary process)

A process $\{Y_t\}$ is *strictly stationary* if the joint distribution of $(Y_{t_1}, \dots, Y_{t_k})$ equals that of $(Y_{t_1+h}, \dots, Y_{t_k+h})$ for all $k \geq 1$, all $(t_1, \dots, t_k)$, and all $h \in \mathbb{Z}$.

Definition 2.2 (Weakly (covariance) stationary process)

A process $\{Y_t\}$ with finite second moments is *weakly stationary* (or second-order stationary, or wide-sense stationary, WSS) if:

1. $\mathbb{E}[Y_t] = \mu$ for all t (constant mean);
2. $\text{Cov}(Y_t, Y_{t+h}) = \gamma(h)$ for all t, h (autocovariance depends only on the lag h , not on t).

Remark

Strict stationarity implies weak stationarity (given finite moments), but not vice versa. For Gaussian processes the two notions coincide. In practice, *weak stationarity* is the working assumption and is almost always what is meant when people say “stationary.”

2.2 Autocovariance and Autocorrelation Functions

Definition 2.3 (Autocovariance function (ACVF))

For a weakly stationary process:

$$\gamma(h) = \text{Cov}(Y_t, Y_{t+h}) = \mathbb{E}[(Y_t - \mu)(Y_{t+h} - \mu)].$$

Properties: $\gamma(0) = \text{Var}(Y_t) \geq 0$; $\gamma(h) = \gamma(-h)$ (symmetry); $|\gamma(h)| \leq \gamma(0)$.

Definition 2.4 (Autocorrelation function (ACF))

$$\rho(h) = \frac{\gamma(h)}{\gamma(0)} \in [-1, 1].$$

The sample ACF at lag h is estimated by

$$\hat{\rho}(h) = \frac{\hat{\gamma}(h)}{\hat{\gamma}(0)}, \quad \hat{\gamma}(h) = \frac{1}{T} \sum_{t=1}^{T-h} (y_t - \bar{y})(y_{t+h} - \bar{y}).$$

Definition 2.5 (Partial autocorrelation function (PACF))

$\alpha(h)$ is the correlation between Y_t and Y_{t+h} after removing the linear effect of the intermediate values $Y_{t+1}, \dots, Y_{t+h-1}$. Equivalently, $\alpha(h)$ is the last coefficient in the best linear predictor of Y_{t+h} from $Y_t, \dots, Y_{t+h-1}$.

The ACF and PACF are the primary diagnostic tools for identifying AR and MA orders:

Process	ACF	PACF
AR(p)	Tails off (exponential or oscillating)	Cuts off after lag p
MA(q)	Cuts off after lag q	Tails off
ARMA(p, q)	Tails off	Tails off

2.3 Testing for Stationarity

2.3.1 Visual inspection

Plot the series, its ACF, and a rolling mean and variance. Non-stationarity often manifests as a trending mean or ACF that decays very slowly.

2.3.2 Augmented Dickey–Fuller test

The null hypothesis is that the series has a *unit root* (is integrated of order 1), i.e. it is non-stationary. The test regresses Δy_t on y_{t-1} and lagged differences:

$$\Delta y_t = \alpha + \beta t + \phi y_{t-1} + \sum_{j=1}^k c_j \Delta y_{t-j} + \varepsilon_t.$$

$H_0 : \phi = 0$ (unit root present). A small p -value rejects the null in favour of stationarity.

2.3.3 KPSS test

Complements the ADF by reversing the hypotheses: the null is *stationarity* and the alternative is a unit root. Using both tests together reduces the risk of misclassification due to low power.

2.4 Achieving Stationarity

Differencing. The first difference $\Delta y_t = y_t - y_{t-1}$ removes a linear trend and transforms a random walk to white noise. The d -th order difference $\Delta^d y_t$ removes a polynomial trend of degree d .

Log transformation. $z_t = \log y_t$ stabilises exponentially growing variance. Combined with differencing: $\Delta \log y_t \approx$ percentage return.

Seasonal differencing. $\Delta_s y_t = y_t - y_{t-s}$ removes a seasonal pattern of period s .

3 Trend, Seasonality, and Classical Decomposition

3.1 Additive vs Multiplicative Decomposition

A time series can be decomposed as:

$$y_t = T_t + S_t + R_t \quad (\text{additive}) \quad \text{or} \quad y_t = T_t \cdot S_t \cdot R_t \quad (\text{multiplicative}),$$

where T_t is the trend-cycle, S_t is the seasonal component (satisfying $\sum_{j=0}^{s-1} S_{t-j} = 0$ in the additive case), and R_t is the remainder.

The multiplicative form is appropriate when the *amplitude* of seasonal swings grows proportionally with the level of the series. Taking logarithms converts a multiplicative decomposition into an additive one.

3.2 Classical (Census) Decomposition

1. Estimate trend $\hat{T}_t$ by a centred moving average of order s :

$$\hat{T}_t = \frac{1}{s} \sum_{j=-(s-1)/2}^{(s-1)/2} y_{t+j} \quad (\text{odd } s).$$

2. De-trend: $d_t = y_t - \hat{T}_t$ (additive).
3. Estimate seasonal factors $\hat{S}_j$ as averages of d_t within each season j .
4. Remainder: $\hat{R}_t = y_t - \hat{T}_t - \hat{S}_t$.

Remark

The classical decomposition has well-known deficiencies: it cannot estimate trend at the endpoints of the series, and it assumes that seasonal patterns are *fixed* over time. STL (Seasonal and Trend decomposition using Loess, Cleveland et al. 1990) addresses both by using local polynomial regression.

Example 3.1 (Monthly Australian Beer Production)

The series 1956–1994 (quarterly, $s = 4$) exhibits a strong seasonal pattern that grows in amplitude over time $\Rightarrow$ multiplicative decomposition is appropriate. After taking logs, additive decomposition gives:

- $\hat{T}_t$: smooth upward trend until mid-1970s, then slight decline.
- $\hat{S}_t$: Q4 is always highest (Christmas); Q1 lowest.

- $\hat{R}_t$: small, mostly random residuals — model fits well.

3.3 Ljung–Box Test for Residual Autocorrelation

After fitting any model, test whether the residuals are white noise:

$$Q(m) = T(T+2) \sum_{h=1}^m \frac{\hat{\rho}_{\varepsilon}(h)^2}{T-h} \stackrel{H_0}{\sim} \chi^2(m-p-q).$$

Rejection (p -value < 0.05) indicates unexplained temporal structure in the residuals.

4 ARIMA Models

4.1 Building Blocks: AR, MA, and ARMA

Definition 4.1 (Autoregressive process AR(p))

$$Y_t = c + \phi_1 Y_{t-1} + \phi_2 Y_{t-2} + \cdots + \phi_p Y_{t-p} + \varepsilon_t, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma^2).$$

Using the *backshift operator* B ($B^k Y_t = Y_{t-k}$):

$$\Phi(B) Y_t = c + \varepsilon_t, \quad \Phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p.$$

The process is *stationary* iff all roots of $\Phi(z) = 0$ lie *outside* the unit circle $|z| > 1$.

Example 4.1 (AR(1))

$Y_t = \phi Y_{t-1} + \varepsilon_t$ with $|\phi| < 1$.

- $\mathbb{E}[Y_t] = 0$, $\text{Var}(Y_t) = \sigma^2 / (1 - \phi^2)$.
- $\rho(h) = \phi^{|h|}$ — exponential decay.
- If $\phi > 0$: positive autocorrelation at all lags (persistence).
- If $\phi < 0$: oscillating autocorrelation (mean-reverting, noisy).
- $\phi = 1$: random walk (non-stationary unit root).

Definition 4.2 (Moving average process MA(q))

$$Y_t = \mu + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \cdots + \theta_q \varepsilon_{t-q}, \quad \varepsilon_t \stackrel{\text{iid}}{\sim} (0, \sigma^2).$$

An MA(q) process is *always stationary*. It is *invertible* (representable as an AR(∞)) iff all roots of $\Theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$ lie outside the unit circle.

Remark

Invertibility is needed to ensure that the “infinite past” of the series recovers the innovations ε_t uniquely — it is the MA analogue of the stationarity condition for AR processes, and is required for unique identifiability of the MA coefficients.

Definition 4.3 (ARMA(p, q) process)

$$\Phi(B) Y_t = c + \Theta(B) \varepsilon_t.$$

Stationary if all roots of $\Phi(z)$ are outside the unit circle; invertible if all roots of $\Theta(z)$ are outside the unit circle.

4.2 Integrated Processes and ARIMA

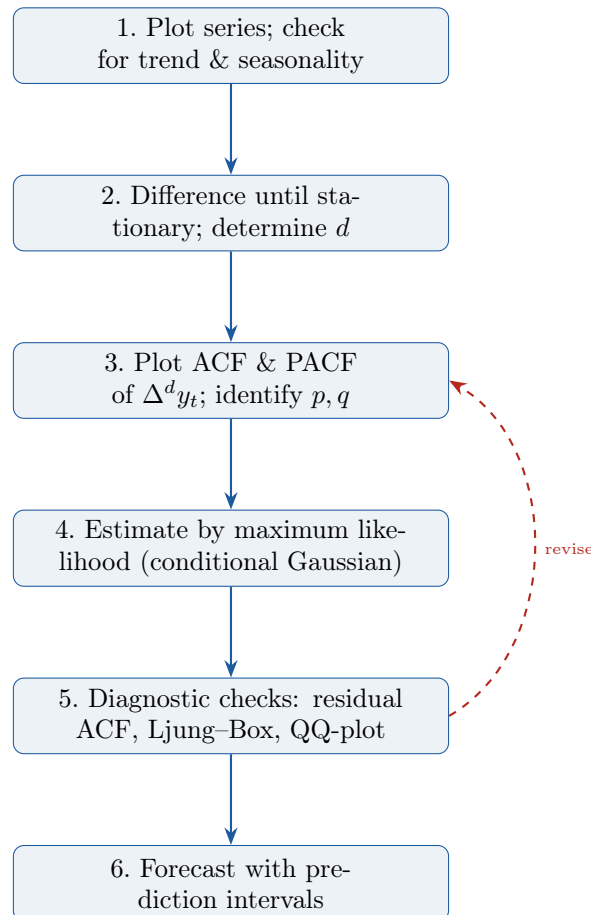
A process Y_t is *integrated of order d* , written $Y_t \sim I(d)$, if $\Delta^d Y_t$ is stationary (and $\Delta^{d-1} Y_t$ is not).

Definition 4.4 (ARIMA(p, d, q) model)

$$\Phi(B) (1 - B)^d Y_t = c + \Theta(B) \varepsilon_t.$$

The three parameters carry distinct roles:

- p — AR order: how many past values enter the model.
- d — integration order: how many times to difference.
- q — MA order: how many past forecast errors enter.

4.3 Model Identification: The Box–Jenkins Workflow

Information criteria for order selection. When PACF/ACF patterns are ambiguous, minimise:

$$\text{AIC} = -2\ell(\hat{\theta}) + 2k, \quad \text{BIC} = -2\ell(\hat{\theta}) + k \log T,$$

where $k = p + q + 1$ is the number of free parameters and $\ell(\hat{\theta})$ is the maximised log-likelihood. BIC penalises complexity more harshly and tends to select sparser models.

4.4 Seasonal ARIMA (SARIMA)

For series with seasonality of period s , introduce seasonal AR and MA operators acting at lag multiples of s :

Definition 4.5 (SARIMA(p, d, q)(P, D, Q) $_s$)

$$\Phi(B) \Phi_s(B^s) (1 - B)^d (1 - B^s)^D Y_t = c + \Theta(B) \Theta_s(B^s) \varepsilon_t,$$

where: $\Phi_s(z^s) = 1 - \Phi_1 z^s - \dots - \Phi_P z^{Ps}$; $\Theta_s(z^s) = 1 + \Theta_1 z^s + \dots + \Theta_Q z^{Qs}$. D is the seasonal differencing order.

Example 4.2 (Monthly airline passengers (Box & Jenkins, 1976))

The famous “airline model” is SARIMA(0, 1, 1)(0, 1, 1) $_{12}$:

$$(1 - B)(1 - B^{12}) \log Y_t = (1 + \theta B)(1 + \Theta B^{12}) \varepsilon_t.$$

First differencing removes the trend; seasonal differencing removes the annual seasonal pattern; the log transform stabilises variance. Typical estimates: $\hat{\theta} \approx -0.40$, $\hat{\Theta} \approx -0.60$.

4.5 Forecasting with ARIMA

The h -step-ahead point forecast $\hat{y}_{T+h|T}$ is the conditional expectation $\mathbb{E}[Y_{T+h} \mid y_1, \dots, y_T]$, computed recursively.

Forecast variance also grows with h . For an ARIMA(p, d, q) model, the forecast error variance increases without bound as $h \rightarrow \infty$ (due to the unit roots), reflecting fundamental irreducible uncertainty.

A $(1 - \alpha)$ prediction interval is:

$$\hat{y}_{T+h|T} \pm z_{\alpha/2} \hat{\sigma}_h,$$

where $\hat{\sigma}_h$ is the estimated standard deviation of the h -step forecast error (computable analytically for ARIMA).

Limitations of ARIMA

ARIMA assumes linearity and Gaussianity. It cannot capture volatility clustering (GARCH-type effects), threshold non-linearities, or very long memory ($d \notin \{0, 1\}$). It also does not easily handle the multivariate setting — that requires VAR.

5 Vector Autoregression (VAR)

5.1 Motivation

In many applications we observe *multiple* time series simultaneously (e.g. GDP, inflation, and unemployment) that influence each other bidirectionally. A univariate ARIMA applied to each series separately ignores these cross-dependencies and loses efficiency.

Definition 5.1 (VAR(p) model)

Let $\mathbf{y}_t = (y_{1t}, \dots, y_{kt})^\top \in \mathbb{R}^k$.

$$\mathbf{y}_t = \mathbf{c} + \mathbf{A}_1 \mathbf{y}_{t-1} + \mathbf{A}_2 \mathbf{y}_{t-2} + \dots + \mathbf{A}_p \mathbf{y}_{t-p} + \boldsymbol{\varepsilon}_t,$$

where $\mathbf{c} \in \mathbb{R}^k$, $\mathbf{A}_j \in \mathbb{R}^{k \times k}$, and $\boldsymbol{\varepsilon}_t \stackrel{\text{iid}}{\sim} (\mathbf{0}, \boldsymbol{\Sigma})$ with $\boldsymbol{\Sigma}$ positive definite.

The VAR(p) contains $k^2 p + k$ free parameters (excluding $\boldsymbol{\Sigma}$) — it grows quadratically in the number of series. With $k = 5$ series and $p = 2$ lags: 55 parameters from T observations. This calls for careful order selection and, in high dimensions, regularisation (e.g. Lasso-VAR).

5.2 Stability Condition

Write the VAR(p) as a VAR(1) in companion form:

$$\mathbf{Y}_t = \mathbf{A} \mathbf{Y}_{t-1} + \mathbf{E}_t, \quad \mathbf{A} = \begin{pmatrix} \mathbf{A}_1 & \mathbf{A}_2 & \dots & \mathbf{A}_p \\ \mathbf{I}_k & \mathbf{0} & \dots & \mathbf{0} \\ \vdots & \ddots & & \vdots \\ \mathbf{0} & \dots & \mathbf{I}_k & \mathbf{0} \end{pmatrix}.$$

The VAR is *stable* (all series stationary) iff all eigenvalues of $\mathbf{A}$ have modulus strictly less than one.

5.3 Estimation

Under stationarity, each equation can be estimated independently by **OLS** — this is efficient because all equations share the same regressors. Maximum likelihood estimation gives the same point estimates and allows joint inference.

5.4 Lag Order Selection

Minimise a multivariate information criterion. For example, the multivariate AIC:

$$\text{AIC}(p) = \log |\hat{\boldsymbol{\Sigma}}_p| + \frac{2k^2 p}{T}.$$

5.5 Granger Causality**Definition 5.2 (Granger causality)**

Series y_{2t} *Granger-causes* y_{1t} if past values of y_{2t} improve the prediction of y_{1t} *beyond* what is already achievable from past values of y_{1t} alone.

In the VAR framework, test $H_0 : A_1^{(12)} = \dots = A_p^{(12)} = 0$ (all off-diagonal coefficients from y_2 in the y_1 equation are zero) using a Wald χ^2 or F -test.

Remark

Granger causality is *predictive*, not *structural*. Finding that y_2 Granger-causes y_1 means past y_2 is useful for predicting y_1 ; it does not imply that y_2 *causes* y_1 in a causal-graph sense.

5.6 Impulse Response Functions

The VMA(∞) representation $\mathbf{y}_t = \sum_{h=0}^{\infty} \Psi_h \boldsymbol{\varepsilon}_{t-h}$ gives the *impulse response matrices* Ψ_h : entry $(\Psi_h)_{ij}$ measures the response of variable i to a unit shock in variable j after h periods.

Example 5.1 (Bivariate VAR for US macroeconomics)

Let $\mathbf{y}_t = (\Delta \log \text{GDP}_t, \Delta \text{CPI}_t)^\top$. A VAR(2) fit to quarterly US data 1960–2019 might reveal:

- GDP growth Granger-causes CPI inflation (demand-pull channel).
- A 1 standard deviation positive shock to GDP growth raises inflation by about 0.15 percentage points over 4 quarters, declining thereafter.
- Variance decomposition: 80% of GDP forecast error at 4 quarters is own-shock; CPI accounts for 20%.

6 State-Space Models and the Kalman Filter

6.1 Motivation

State-space models provide a unified framework for:

- **Structural time series**: explicitly decompose into trend, seasonal, and irregular components with probabilistic semantics.
- **Noisy observations**: the latent state is observed through a measurement equation corrupted by noise.
- **Missing data, mixed frequencies**: handled naturally.
- **Multivariate and nonlinear extensions** (EKF, UKF, particle filters).

6.2 Linear Gaussian State-Space Model

Definition 6.1 (Linear Gaussian state-space model)

$$\text{Measurement equation: } \mathbf{y}_t = \mathbf{H}\mathbf{x}_t + \mathbf{v}_t, \quad \mathbf{v}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{R}), \quad (1)$$

$$\text{Transition equation: } \mathbf{x}_t = \mathbf{F}\mathbf{x}_{t-1} + \mathbf{w}_t, \quad \mathbf{w}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (2)$$

where $\mathbf{x}_t \in \mathbb{R}^m$ is the *latent state*, $\mathbf{y}_t \in \mathbb{R}^k$ is observed, $\mathbf{H} \in \mathbb{R}^{k \times m}$ is the observation matrix, $\mathbf{F} \in \mathbb{R}^{m \times m}$ is the transition matrix, $\mathbf{R}$ is observation noise covariance, and $\mathbf{Q}$ is process noise covariance. The noise sequences are mutually independent.

6.3 The Kalman Filter

The Kalman filter is the recursive algorithm that computes, in closed form, the optimal (minimum mean-squared-error) linear estimate of the state given all observations up to time t .

Denote: $\mathbf{x}_{t|s} = \mathbb{E}[\mathbf{x}_t \mid y_1, \dots, y_s]$ (state estimate); $\mathbf{P}_{t|s} = \text{Cov}(\mathbf{x}_t - \mathbf{x}_{t|s})$ (estimation error covariance).

Algorithm 1: Kalman Filter

Input: Initial state $\mathbf{x}_{0|0}$, covariance $\mathbf{P}_{0|0}$, matrices $\mathbf{F}, \mathbf{H}, \mathbf{Q}, \mathbf{R}$

```

1 for  $t = 1, 2, \dots, T$  do
2   Predict (time update);
3    $\mathbf{x}_{t|t-1} = \mathbf{F}\mathbf{x}_{t-1|t-1}$ ;
4    $\mathbf{P}_{t|t-1} = \mathbf{F}\mathbf{P}_{t-1|t-1}\mathbf{F}^\top + \mathbf{Q}$ ;
5   Observe innovation and its covariance;
6    $\mathbf{v}_t = \mathbf{y}_t - \mathbf{H}\mathbf{x}_{t|t-1}$  # innovation (prediction error)
7    $\mathbf{S}_t = \mathbf{H}\mathbf{P}_{t|t-1}\mathbf{H}^\top + \mathbf{R}$ ;
8   Compute Kalman gain;
9    $\mathbf{K}_t = \mathbf{P}_{t|t-1}\mathbf{H}^\top\mathbf{S}_t^{-1}$ ;
10  Update (measurement update);
11   $\mathbf{x}_{t|t} = \mathbf{x}_{t|t-1} + \mathbf{K}_t\mathbf{v}_t$ ;
12   $\mathbf{P}_{t|t} = (\mathbf{I} - \mathbf{K}_t\mathbf{H})\mathbf{P}_{t|t-1}$ ;
13 end

```

6.3.1 Intuition for the Kalman gain

$\mathbf{K}_t$ interpolates between two extreme cases:

- $\mathbf{R} \rightarrow \infty$ (very noisy observations): $\mathbf{K}_t \rightarrow \mathbf{0}$, so the filter ignores new data and relies on the prior prediction.
- $\mathbf{Q} \rightarrow \mathbf{0}$ (deterministic transitions): $\mathbf{K}_t \rightarrow \mathbf{H}^{-1}$ (if invertible), so the filter fully trusts the observation.

6.3.2 Log-likelihood via the innovation form

Since innovations are Gaussian:

$$\ell = -\frac{1}{2} \sum_{t=1}^T \left[\log |\mathbf{S}_t| + \mathbf{v}_t^\top \mathbf{S}_t^{-1} \mathbf{v}_t \right] + \text{const.}$$

This allows estimation of unknown parameters $(\mathbf{F}, \mathbf{H}, \mathbf{Q}, \mathbf{R})$ by maximum likelihood via numerical optimisation.

6.4 Encoding Classical Models in State-Space Form

Local level model (random walk + noise).

$$y_t = x_t + v_t, \quad v_t \sim \mathcal{N}(0, r); \quad x_t = x_{t-1} + w_t, \quad w_t \sim \mathcal{N}(0, q).$$

$F = H = 1$. Signal-to-noise ratio $\lambda = q/r$ determines how quickly the filter adapts to new levels.

ARIMA in state-space. Any ARIMA model has an exact state-space representation, so the Kalman filter provides an alternative (and numerically stable) method for computing the likelihood of an ARIMA model.

Structural time series. Trend, seasonal, and irregular components can each be modelled as a separate state, then combined. For example:

$$y_t = \mu_t + \gamma_t + \varepsilon_t,$$

where $\mu_t = \mu_{t-1} + \nu_t$ (stochastic trend) and $\gamma_t = -\sum_{j=1}^{s-1} \gamma_{t-j} + \omega_t$ (stochastic seasonal).

Example 6.1 (Tracking a noisy sensor)

A temperature sensor reads $y_t = x_t + v_t$ where x_t (true temperature) follows a slow random walk $x_t = x_{t-1} + w_t$. Calibration data give $r = 1^\circ\text{C}^2$, $q = 0.01^\circ\text{C}^2$. The Kalman filter maintains a running estimate that is much less noisy than the raw sensor, with the gain K converging to the steady-state value $K^* = \sqrt{q(q+r)^{-1}} \approx 0.10$ — only 10% weight on each new observation.

6.5 Extensions

- **Kalman smoother (RTS smoother):** given *all* observations $y_1, \dots, y_T$, compute $\mathbf{x}_{t|T}$ retroactively — optimal for offline signal extraction.
- **Extended Kalman Filter (EKF):** linearises nonlinear transition/observation equations via first-order Taylor expansion.
- **Unscented Kalman Filter (UKF):** propagates a set of sigma points through the nonlinear function — more accurate than EKF for moderately nonlinear systems.
- **Particle Filter:** Monte Carlo approximation for fully nonlinear, non-Gaussian state-space models.

7 Summary and Connections

Model	Data type	Stationarity required?	Multivariate?	Key assumption
AR(p)	Univariate	Yes	No	Linearity, Gaussianity
MA(q)	Univariate	Yes	No	Linearity, Gaussianity
ARIMA(p, d, q)	Univariate	After differencing	No	Linearity
SARIMA	Univariate	After differencing	No	Fixed seasonal period
VAR(p)	Multivariate	Yes	Yes	Linearity, no structural break
State-space (Kalman)	Uni- or multi.	No (latent state)	Yes	Linearity, Gaussianity

Key Idea

All classical models in this section share a common **linearity** assumption: the future is a linear combination of the past. Part II of these lecture notes replaces this assumption with nonlinear function approximators — recurrent and attention-based neural networks — which can in principle capture arbitrarily complex temporal dependencies, at the cost of interpretability and data efficiency.

A Key Formulae Reference

A.1 ACF of standard processes

$$\text{AR}(1): \quad \rho(h) = \phi^h, \quad h \geq 0$$

$$\text{MA}(1): \quad \rho(1) = \frac{\theta}{1 + \theta^2}, \quad \rho(h) = 0 \text{ for } h \geq 2$$

$$\text{ARMA}(1,1): \quad \rho(1) = \frac{(1 + \phi\theta)(\phi + \theta)}{1 + 2\phi\theta + \theta^2}, \quad \rho(h) = \phi \rho(h-1), \quad h \geq 2$$

A.2 Kalman steady-state

For the scalar local level model with $F = H = 1$:

$$P^* = \frac{q}{2} \left(1 + \sqrt{1 + 4r/q} \right), \quad K^* = \frac{P^*}{P^* + r}.$$

A.3 AIC / BIC quick reference

$$\text{AIC} = 2k - 2 \ln \hat{L}, \quad \text{BIC} = k \ln T - 2 \ln \hat{L}.$$

Both penalise additional parameters; BIC penalises more for $T > 7$.